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CS-158A Computer Networks

Lab 1

Checkpoint 1:

The request returned status code 200, which means the request was completed without issue

```
cpetri@afterlife:~$ curl -v --http1.1 http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file1.html
* Host gaia.cs.umass.edu:80 was resolved.
* IPv6: (none)
* IPv4: 128.119.245.12
*   Trying 128.119.245.12:80...
* Established connection to gaia.cs.umass.edu (128.119.245.12 port 80) from 192.168.0.32 port 44422
* using HTTP/1.x
> GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1
> Host: gaia.cs.umass.edu
> User-Agent: curl/8.18.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Wed, 02 Sep 2026 20:38:27 GMT
< Server: Apache/2.4.62 (AlmaLinux) OpenSSL/3.5.5 mod_fcgid/2.3.9 mod_perl/2.0.12 Perl/v5.32.1
< Last-Modified: Tue, 28 Oct 2025 05:59:01 GMT
< ETag: "80-64231b6717e87"
< Accept-Ranges: bytes
< Content-Length: 128
< Content-Type: text/html; charset=UTF-8
<
<html>
Congratulations. You've downloaded the file
http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file1.html!
</html>
* Connection #0 to host gaia.cs.umass.edu:80 left intact
```

Figure 1: Code 200 returned with descriptor OK following it

Checkpoint 2:

HTTP 1.0 has a full 3 way handshake for each request. 1.1 keeps the connection open for a time to minimize handshake overhead.

HTTP 1.0

2.621641011	192.168.0.32	128.119.245.12	HTTP	187 GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.0
2.706162931	128.119.245.12	192.168.0.32	TCP	66 80 → 37072 [ACK] Seq=1 Ack=122 Win=65152 Len=0 TSval=2051
2.707812413	128.119.245.12	192.168.0.32	HTTP	518 HTTP/1.1 200 OK (text/html)

Figure 2: 2.707812413 - 2.621641011 = 0.086171402 s

HTTP 1.1

11 1.820876698	192.168.0.32	128.119.245.12	HTTP	187 GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1
12 1.853134095	192.168.0.32	162.159.135.234	TLSv1.2	187 Application Data
13 1.865215256	162.159.135.234	192.168.0.32	TCP	66 443 → 56374 [ACK] Seq=87 Ack=122 Win=17 Len=0 TSval=20016
14 1.904885297	128.119.245.12	192.168.0.32	TCP	66 80 → 60482 [ACK] Seq=1 Ack=122 Win=65152 Len=0 TSval=2057
15 1.904886239	128.119.245.12	192.168.0.32	HTTP	499 HTTP/1.1 200 OK (text/html)

Figure 3: 1.904886239 - 1.820876698 = 0.084009541 s

Checkpoint 3:

The IP address, URL, and a description of the client hello is still visible. The return code and contents are hidden from us when using HTTPS.

9	2.174865099	192.168.0.32	128.119.245.12	TCP	1514 37864 → 443 [ACK] Seq=1 Ack=1 Win=64512 Len=
10	2.174871260	192.168.0.32	128.119.245.12	TLSv1.3	199 Client Hello (SNI=gaia.cs.umass.edu)
11	2.249831348	162.159.135.234	192.168.0.32	TLSv1.2	292 Application Data

Figure 4: URL and IP address visible

Apply a display filter ... <Ctrl-/>						
Packet list		Display filter		https		
Options: Narrow Wide Case sensitive Backwards Multiple occurrences						
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	TPLink_58:62:2b	Broadcast	ARP	42	Who has 192.168.0.172? Tell 192.168.0.172
2	1.016972138	TPLink_58:62:2b	Broadcast	ARP	42	Who has 192.168.0.172? Tell 192.168.0.172

Figure 5: No HTTP requests